

# Accuracy and reliability of self-reported data in interorganizational networks \*

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Although network analysis is becoming a more viable and informative methodology for the study of interorganizational behavior, issues involving the accuracy and reliability of self-reported or 'cognitive' network data are still unresolved. Two networks based on information and coordination relationships in each of four major cities are examined. Reliability of responses is assessed as the percent of confirmed relationships between agency respondents and by correlating the extent of agreement in each network. In addition, two possible measurement errors are examined: systematic errors which are network-independent and idiosyncratic errors which are network-dependent. These errors are assessed through correlational analyses of the intensity of an organization's relationships with the number of its relationships that are confirmed in each network. Also systematic errors are accounted for by calculating the proportion of confirmed relationships within the different response categories used to judge the strength of relationship between ego organizations in the network. It is concluded that boundary spanning personnel of organizations can provide fairly reliable responses to network generated questions, but that errors in the data tend to be systematic in the way that prior studies of interpersonal cognitive networks report.

## 1. Introduction

Over the past two decades there has been a rapid increase in the use of network analysis techniques in interorganizational studies (Oliver

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1990). Spurred on by recent advances in methodology and mathematical sophistication, network analysis is becoming a well-grounded method for studying organizational behavior (Mizruchi and Schwartz 1987). However, caution has been recommended in its use due to important and still unresolved issues concerning the reliability of the data (Marsden 1990). For interorganizational (IO) studies these issues loom even larger given the absence of any systematic work investigating these concerns. At this point IO network analyses progress under the assumption that the errors in and the reliability of the data are comparable with those in traditional survey research. This paper will focus on two potential sources of reliability problems in network data collected on organizational relationships: the reliability of responses from organizational respondents concerning the presence/absence of relationships and the extent of systematic errors in those data.<sup>1</sup> It is not difficult to see that either or both of these problems, if present, could jeopardize the use of network data in IO studies. An investigation of these issues provides some indication about the quality of the network data collected. We will first discuss, with a focus on data quality issues, the relevance of using network data in the field of IO analyses. Following is an analysis of these data reliability issues using recently collected IO network data. We conclude by assessing the quality of the data in terms of the findings.<sup>2</sup>

The strengths of network data in the field of IO studies are twofold. First, network methodology utilizes data in a relational format, such that the sum total of organizational relations in a well-delineated system provides a clear and straightforward way of operationalizing the environment in which organizations function. Second, because the methodology provides a way of understanding the organizational environment as an epiphenomenon, that is being something more than the sum total of the organizations and their attributes, the method is

<sup>1</sup> Regrettably, the data used are not ideal for a definitive analysis of these issues. However, the data do allow for an important initial examination of these issues. Any conclusions should be tempered by the judged ability of the data to address these issues.

<sup>2</sup> Hereafter we equate reliability not, as it is traditionally used in the measurement literature, as the extent to which a measure when repeated provides consistent measurement, but as the extent to which relationships are confirmed by organizational respondents as either being present or absent. In this sense it is more of an indicant of quality.

easily suited for use within established sociological theories (see Burt 1982; Granovetter 1979; Blau 1982). Third, network methods can locate organizations within well-defined environmental positions (Morrissey *et al.* 1985) allowing researchers a way of ascribing positional level attributes to individual organizations or providing a way of moving the unit of analysis from individual organizations to a more emergent level (McPherson 1982). Factors operating at a systems level can then be entered into equations of organizational behavior along with the specific organizational attributes (e.g. size, resources, age, etc.) in order to estimate the interplay between social structure and organizational attributes.

Each of these strengths point to network data being able to provide measures of an environment through the analyses of empirical links, ties, or exchanges among social positions. This structure, the *sine qua non* of patterned relationships, is emergent and not reducible to the individual properties of organizations (Durkheim 1947; Warren *et al.* 1971; Van de Ven and Ferry 1980). These data do not represent yet another way of viewing organizational phenomena from organizational attributes: these emergent phenomena must be estimated separately from organizational attributes (Wellman 1988). Detecting this structural phenomenon through the relationships of a given set of organizations is what the mathematics of network analyses are about (Burt and Minor 1983; Burt 1976; Breiger *et al.* 1975; Doreian 1988; Faust 1988; White *et al.* 1976; Sailer 1978).

Although analysis of network data depends upon the quantification of relationships, it is still unclear as to the most objective or precise way to obtain information about them. One strategy used both within and outside of IO studies has been to use proxies of actual relationships. This method relies on secondary data to provide accounts of actual exchanges between organizations. Searches of secondary and archival data (Lievrouw *et al.* 1987; White and McCann 1988) can provide reasonable indicators of relationships. Some examples of this type of accounting within IO studies are interlocking directorates (Pennings 1980; Allen 1982; Mintz and Schwartz 1985; Burt 1983; Koenig *et al.* 1979). A second and quite different method used to gather reliable relational data involves direct observation by the researcher, usually in small group settings (Foster and Seidman 1984; Bernard and Killworth 1977). To our knowledge, this strategy has not been used in IO studies.

There are a couple of problems with using either indicators or direct observations as a method of collecting network data in IO studies. In the case of using direct observational data, the size of the networks must be small and manageable indicating this method may only be practical in terms of intraorganizational studies. One would also be hard pressed to find a relationship that could be observed taking place between organizations. While the data collected in this fashion may circumvent the problem of reliable data, documenting and counting relationships in large networks is problematic since IO networks probably, depending on the nature of the study, include many organizations in an environment just too large to observe. Further, in direct observational studies involving IO relations, decisions must be made as to when a relationship actually begins and ends; a problematic yet crucial aspect of any network analysis strategy.

Although the size of networks can typically be larger when using secondary data than with direct observation, this method also has problems. Documenting relationships is not as much of a problem as the concern over properly defined samples. For example, depending on the completeness of the source data, some organizations may be excluded from the network since there are no data present on these organizations. If these are necessary for inclusion in the network, accurately defining the bounds of the network is not possible which renders any analysis questionable (Laumann *et al.* 1983).

In other studies, network data are collected on IO relations through the use of surveys and questionnaires (Morrissey *et al.* 1985; Bolland 1985). Rather than relying on quantitative counts of exchanges, these methods rely on a respondent's knowledge of the links between their organization and all others in the network. Survey methods seem well suited to IO network data collection in several ways. First, they provide a way to a priori and correctly bound the network.<sup>3</sup> As a result all necessary organizations are included in the survey design. Secondly, the issue of what constitutes a link can be defined in the survey and more easily measured. Furthermore, the design allows for larger networks in terms of the number of organizations studied.<sup>4</sup>

<sup>3</sup> For a good exposé of this issue and one that provides a caveat to our statement, see Doreian and Woodard (1992).

<sup>4</sup> Including organizations is not a problem in collecting survey network data. However, keeping the respondents' attention and focus on a large number of organizations is a potential problem.

Because of these strengths, this method seems promising, but is not without its problems. Studies have shown, for example, that people often have difficulty identifying who they have interacted with over specific time periods (see Killworth and Bernard 1979). Since surveys rely on individual memories, the quality of the data can be suspect on the grounds of reliability.

Recently, Marsden (1990: 445–146) summarized the problems of reliability associated with self-reported or cognitive network data, noting that few studies have dealt with this subject and calling for more assessments of the reliability of ‘cognitive networks’. Initial results suggest that cognitive network data are subject to systematic rather than random errors, with persons involved in stronger relationships being more likely to confirm their interaction patterns (Romney and Weller 1984; Hammer 1985). This is encouraging for studies of open-ended networks using name generators and snowball sampling techniques which inherently include strong ties, but may be more troubling for studies of networks that preclude a broader range of interactions. For IO studies, the problem is compounded (Marsden 1990: 444) given the absence of ongoing studies investigating the quality of network data collected from agents representing organizations. There is, however, a promising note. Organizational links over time can become quite routine and research has shown that response reliability of the more institutionalized relationships might be higher than those relationships typically studied in interpersonal networks (e.g. friendship or advice) which tend to be more dynamic. Respondents are more likely to accurately recall typical and routine relations as opposed to relationships that refer to specific time and/or content references (Marsden 1990: 447; Freeman *et al.* 1987). This may be good news to researchers who want to employ network analysis techniques in their IO studies assuming that the organizations have routinized the relations of interest.

In this paper we test the reliability of, and the extent of systematic errors in, recently gathered IO network data collected from organizational respondents. We do this through analyzing the amount of confirmed, *mutual* acknowledgments of relationships. Because of the difficulty involved in objectively confirming or monitoring self-reported ties in large groups, “one alternative method is to presume that *mutually* acknowledged ties are genuinely present and to see how often citations are reciprocated” (Marsden 1990: 448).

## 2. Data collection

The data used in this paper were gathered in four US metropolitan cities during the period May–August 1989 as part of an ongoing evaluation of the Robert Wood Johnson (RWJ) Foundation Program on Mental Illness (Shore and Cohen 1990; Goldman *et al.* 1990). The sites included from 31 to 44 organizations that provide mental health and other community support services to persons with a chronic or serious and persistent mental illness. Data collection procedures followed those employed in prior studies (Morrissey *et al.* 1985; Morrissey and Schott 1990). The organizations were identified in consultation with staff from the local (RWJ) demonstration projects, from conversations with organization directors and other persons knowledgeable about these issues, and from a directory of local mental health and psychosocial support services.

Data about the formal network of services at each site were gathered from a key informant (or ‘boundary spanner’) in each organization. Boundary spanning roles (Aldrich and Herker 1977; Tushman and Scanlan 1981) are those occupied by persons who serve as liaisons with other organizations or who are deemed as particularly knowledgeable about the organization’s relationships (e.g. agency, unit, or program director). The respondents chosen were identified from interviews conducted with representatives of the major organizations, or recommendations made by RWJ project staff, or direct inquiries with organization directors.

Data collection involved a three step process. First, the site coordinator phoned each organization respondent to explain the study’s purpose, to request their participation, and to schedule an interview. Second, an in-person interview was conducted with the respondent. The semi-structured interview lasted approximately 45 minutes and dealt with several aspects of the respondent’s job and organization. Following the interview, the interviewer oriented the respondent to the format and content of a self-administered questionnaire containing 16 questions about the relationships between the respondent’s organization and each of the other organizations in the community support system. Ultimately, a response rate of 100% for all organizations in all four cities was obtained.

Although the study design was not specifically developed to address these data reliability issues, the data have several attributes which

make them appropriate for examining the issues. First, the networks are relatively large in size when compared to many network studies. In these four sites the number of possible dyadic relationships range from 930 to 1892, and when the data are pooled across sites the total number of possible relationships is 5710. This allows for a more systematic assessment of network structure. Second, the networks represented a well-bounded study design that explores relationships identified by key informants among all organizations that deal with mental health services to persons with a chronic mental illness. Although bounding networks is an all together separate issue from exploring data reliability (Laumann *et al.* 1983; Doreian and Woodard 1992), it impinges on this study to the extent that when considering all sites, response correlations are not biased by the inclusion or exclusion of organizations that for the most part are strongly connected. Third, the networks were bounded a priori, avoiding the problem of name generated studies that limit the size of the networks. This feature also controls for the bias associated with name generated studies where an actor's inclusion by definition represents a very strong tie. Last, for each site there are specific questions that allow for the investigation of mutual citation of relationships. The reciprocal nature of these questions can provide clues as to the quality of responses in the self-reported relationships by organizational representatives.

### 3. Measurement

In order to assess the accuracy of the organizational respondent's data, two relational network questions from the survey are used. The first asks respondents: "To what extent does your agency *receive information* for coordination, control, planning, or evaluation purposes from this agency?" A second asks: "How well *coordinated* are the activities of your agency with those of this other agency?" Respondents answered each question on a Likert scale of 0 to 4 where 0 signified 'not at all' and 4 signified 'a lot'. These are questions for which symmetrical relations can be reasonably assumed and they

provide data from which to assess mutually cited relationships.<sup>5</sup> First, we will investigate the response reliability of the organizational respondents and second explore the issue of systematic errors in network data.

#### 4. Reliability using organizational agents

Confirmed presence or absence of links is used to estimate the reliability or quality of the network data provided by organizational representatives. Bernard *et al.* (1982: 32) discussed two separate aspects of data quality in informant responses; having an accurate knowledge of who they had communicated with, and the ability to correctly recall the intensity of the communication. We will consider both of these aspects in turn. An initial and straightforward way of assessing data quality is to investigate the reciprocity of relationships by correlating the responses from organization  $i$  to  $j$ , with organization  $j$ 's responses to  $i$ . This is accomplished by creating an adjacency matrix where the original 0 to 4 data are dichotomized. Dichotomizing the data allows us to correlate the confirmation of relationships only, it does not require that the respondents accurately confirm the intensity of the relation. Although any cutpoint chosen to dichotomize the data is somewhat arbitrary, it seemed reasonable that the most appropriate cutpoint is where responses greater than 0 are set equal to 1; with 1 representing the presence of a relation. By transposing this adjacency ( $i$  to  $j$ ) matrix, one additional matrix ( $j$  to  $i$ ) is created. An initial assessment of the reliability of the IO network responses is accomplished by correlating the adjacency and transposed matrices (for each question). The results are shown in Table 1.<sup>6</sup>

<sup>5</sup> We also compared the results with a question that is assumed not to be symmetrical: "To what extent does your agency receive clients from this other agency?" The correlations on these client exchange data for the four cities ranged from 0.08 to 0.042 with the average correlation being 0.062. These results suggest that information is much more symmetrical in nature than we supposed, given that the correlations for client referrals are at least 7 times lower than information and coordination. Although we had some concern that information may not be symmetrical we included it in the analysis for two reasons. First, the results for coordination and information were so similar we were confident that both relations were symmetrical in nature, and second two questions rather than one provide more data for conclusions.

<sup>6</sup> The analysis is implemented through the correlation option under UCINET's MATSIM procedure (MacEvoy and Freeman 1987).



Table 1

Correlation of reported relations between organizations for two networks

|             | N  | Information network |                    | Coordination network |                    |
|-------------|----|---------------------|--------------------|----------------------|--------------------|
|             |    | Density             | Pearson's <i>r</i> | Density              | Pearson's <i>r</i> |
| City 1      | 39 | 0.37                | 0.336              | 0.39                 | 0.384              |
| City 2      | 38 | 0.41                | 0.398              | 0.41                 | 0.409              |
| City 3      | 44 | 0.38                | 0.453              | 0.34                 | 0.426              |
| City 4      | 31 | 0.43                | 0.587              | 0.43                 | 0.497              |
| Pooled avg. |    | 0.41                | 0.44               | 0.40                 | 0.43               |

Table 1 shows the correlations for each city and network as determined by the extent of confirmed responses. Larger correlation coefficients indicate higher numbers of confirmed relationships, and thus higher data reliability. Based on the pooled averages, correlations of 0.44 for information and 0.43 for coordination were obtained. At first glance the correlations seem small given the supposed reciprocal nature of the questions. In fact, the correlations may be small for the simple reason that the questions are not as symmetrical as we assumed.

It is also interesting to examine the pattern of correlations and percentages of confirmed relationships across the four cities. The correlation patterns are very similar across networks within cities as shown by the rank order (Table 1). Overall, in City 1 the organization's representatives tended to have slightly lower confirmed responses than the other cities. In City 4 the responses have the most confirmation.<sup>7</sup> Any variation in the pattern of confirmed responses may be a function of the network densities where by chance more dense networks will have higher confirmation percentages. Table 1 also presents these densities. In these data, variations are probably not due to densities since there is little difference either across questions or across cities. The largest density differences are 6 percentage points between City 1 and City 4 in the information network, and 9 percent-

<sup>7</sup> This maybe due to sample variability either with regard to respondent samples, or organizational samples (or both). In both cases the respondent representing the organization and the organization itself were chosen using stringent criteria for inclusion and the samples were finalized with comparability in mind.

Table 2

Correlation of original matrices and their transposes

| City        | <i>N</i> | Information network | Coordination network |
|-------------|----------|---------------------|----------------------|
| City 1      | 39       | 0.331               | 0.444                |
| City 2      | 38       | 0.492               | 0.463                |
| City 3      | 44       | 0.452               | 0.385                |
| City 4      | 31       | 0.601               | 0.484                |
| Pooled avg. |          | 0.47                | 0.44                 |

age points between City 3 and City 4 in the coordination network.<sup>8</sup> To our thinking these similar confirmation patterns are encouraging and may signal that organizational representatives (if carefully chosen) can provide data that are consistent across network domains.

Second, we correlated the original matrix with its transpose which includes both the presence and intensity of a relation between pairs of organizations. Table 2 present these results. The similarity of the results in Tables 1 and 2 reveal that the intensity of the relationship is as likely to be accurately recalled as the presence of the relationship. This is true for both networks. Whereas respondents are only slightly more aware of relationships in the coordination network, they are slightly less accurate in their estimation of the intensity of the relationship.<sup>9</sup>

Unfortunately, we have no benchmark numbers with which to compare the findings. We can, however, compare these findings to results obtained in individual ego-centered networks by examining the actual percentage of agreement on an organization by organization basis. Table 3 presents these data. Overall, the average percent of agreement was 70% for information and 68% for coordination. For both questions then, approximately seven out of ten relationships

<sup>8</sup> However, hinting that more dense networks have a higher rate of confirmed relations, the highest and lowest densities in the information network do have the highest and lowest correlations respectively. The mid range density values do not follow this pattern, nor is the pattern found to be true in the coordination network. However, with eight cases no concrete conclusions can be made.

<sup>9</sup> We compared our findings with the correlations generated on four separate studies conducted by Bernard and colleagues. These data are included and described within the UCINET 4.0 program (Borgatti, Everett and Freeman 1992). The correlations for these studies ranged from 0.37 to 0.53. Two of the correlations are negative.

Table 3

Percentage of confirmed relationships and the correlation of agreements in information and coordination networks

|                    | <i>N</i> | Information network (%) | Coordination network (%) | Pearson's <i>r</i> |
|--------------------|----------|-------------------------|--------------------------|--------------------|
| City 1             | 39       | 71.9                    | 72.0                     | 0.276              |
| City 2             | 38       | 66.9                    | 66.8                     | 0.219              |
| City 3             | 44       | 69.0                    | 67.3                     | 0.098              |
| City 4             | 31       | 72.3                    | 67.3                     | 0.317              |
| Overall percentage | 152      | 70.0                    | 68.4                     |                    |

cited by either respondent were confirmed. Within ego-centered networks where data are collected on relationships among best friends, percentages of confirmed responses hover around 75% across different networks (see Marsden 1990: 450).<sup>10</sup> The confirmation percentages found in our IO network data appear similar to these figures although our percentages are slightly lower. However, when compared with the results published by Bernard and colleagues (1982: 44), where only approximately 50% of the respondents were able to accurately recall those with whom they had recently communicated via electronic mail, and 52% of respondents who were able to identify session participants as reported by Freeman and associates (1987a: 312), these results are slightly more encouraging. In sum, the reliability of the data as reported by the organizational representatives in our study appears to be, at the least, comparable to egocentric networks and more acceptable than those reported in cognitive communication networks.

The consistency shown in Tables 1, 2, and 3 raises another issue that Laumann and Knoke (1986) have noted namely, that different networks should generate different patterns of response. Given the high percentage of agreements on both questions as shown above, high correlations across networks might suggest that the responses are not network specific, but more likely a general organizational orientation or respondent predisposition. For the two different networks (with some domain similarity) we investigate the organizational pat-

<sup>10</sup> The confirmation percentages used in the ego-centered studies were the number of persons cited as being part of a relation at two points in time, whereas the confirmation percentages in this study were the number of relations that were confirmed for *i* and *j*. Because the design of the studies are so dissimilar comparisons should be made with caution.

terns across networks. To do this, the percentage of agreement for each organization on both network questions is correlated. Table 3 provides the correlations. None of the correlations are significant at the 0.05 level and there appears to be no real pattern of responses across networks. In each city then, organizational respondents appear to be providing reliable as well as network specific responses.

## 5. Systematic error in cognitive network data

Given that the above analysis suggests that the network data are relatively reliable, we next investigate the extent to which those organizations sharing strong relations influence the informants' perceptions causing them to be more aware of, and thus more likely to report these relationships. If so, this would introduce systematic errors in network data. The hypothesis being tested is that these strongly related organizations will have higher numbers of confirmed relationships than those weakly linked as a function of the strength of the relationships. Rather than correlating the entire network of scores, a score for each organization is computed indicating the number of relations that were confirmed and correlated with the percent of their relations to the other organizations that are particularly strong. If systematic error based on the content of the relationship is apparent in the networks, there will be high correlations between these two scores for those organizations with a high percentage of their relations strong and a high percentage of their relations confirmed.

Computing a confirmation score required transposing the 0,1 adjacency matrix of reported relationships and comparing the  $i$  to  $j$  responses in both matrices. For each organization, a total was taken on the number of times a confirmation occurred (*i.e.*  $i, j = j, i = 1$ ). The resultant column vector contains a sum of the number of times an organization reported a confirmed relationship. This sum total was then divided by the number of relationships the respondent for organization  $i$  reported. Similarly another vector of scores, representing the percentage of an organization's relationships that are strong, required building an adjacency matrix from the original response matrix where responses 3 or 4 are set equal to 1, where 1 represents a strong relation, and all other responses are set to 0. A sum total is then taken across row  $i$ , and is divided by the number of relationships

the respondent for organization  $i$  reported.<sup>11</sup> This proportion is the number of an organization's mutual relations that are strong compared to their total number of relations. Table 4 provides the descriptive statistics on these two variables.

Interestingly, within each city there is a pattern of fairly consistent mean responses for both information and coordination questions. On average, approximately 23% of the relations in the information networks and 28% in the coordination networks are considered strong. The respondents in City 4 tend to report fewer strong ties, for example, only 17% of the relationships on average in the information network are considered strong ties. In viewing the number of reciprocated relations, on average 63% of all relations mentioned are confirmed by both organizations across the two networks. In City 1, on average, 74% of the relationships in the coordination network were confirmed. For the most part the relationships in the coordination network tend to have a higher percent of confirmation. In fact in all cities except City 4 (for both networks), many respondents had all of their relationships confirmed. For the most part these organizations with all their relations confirmed are not well connected in the network having on average 4 or 5 relationships. However, some respondents reported as many as 19 relationships all of which were confirmed. These results may be interpreted as following the response bias described by Freeman and associates (1987a: 314–315) where respondents who are increasingly experienced or central to a network will respond in accordance to a mental structure based on more long-term, typical situations. If this is the case, the respondents of more central organizations will, by default, include organizations routinely into their ego networks that in reality they have no relation with. However, because of the nearly 70% confirmation rate overall in these IO networks, we agree with their conclusion that these type of structurally perceived relations “will introduce very few errors” (Freeman *et al.* 1987a: 315). Again, in Table 4, we were pleased to see the consistent response patterns across the two networks and four cities.

By correlating the percentage of an organization's relations that are strong with the percentage of its relations that were confirmed, we can address whether systematic errors based on the nature of the relation-

<sup>11</sup> By dividing the sum total by the number of relationships reported rather than the total possible number of relationships we are, in effect, discounting no relationship as a valid choice.

Table 4  
Descriptive statistics on two created variables: the percent of strong relations and the number of links that were confirmed in information and coordination networks

|         | Per cent of strong ties |      |      |                      |      | Per cent of confirmed ties |     |      |                      |      |
|---------|-------------------------|------|------|----------------------|------|----------------------------|-----|------|----------------------|------|
|         | Information network     |      |      | Coordination network |      | Information network        |     |      | Coordination network |      |
|         | Min                     | Max  | Mean | Min                  | Max  | Min                        | Max | Mean | Min                  | Max  |
| City 1  | 0                       | 87.0 | 21.6 | 0                    | 88.9 | 0                          | 100 | 59.6 | 33.3                 | 100  |
| City 2  | 0                       | 100  | 24.8 | 0                    | 100  | 0                          | 100 | 63.4 | 0                    | 100  |
| City 3  | 0                       | 100  | 27.7 | 0                    | 100  | 0                          | 100 | 59.1 | 0                    | 100  |
| City 4  | 0                       | 58.3 | 17.0 | 0                    | 66.7 | 0                          | 100 | 64.7 | 0                    | 91.0 |
| Overall |                         |      | 22.8 |                      |      |                            |     | 61.7 |                      | 66.4 |

Table 5

Correlation between the percent of strong relations and the number of confirmed links in information and coordination networks

| City   | Number of organizations | Number of dyads | Information network | Coordination network |
|--------|-------------------------|-----------------|---------------------|----------------------|
| City 1 | 39                      | 1482            | -0.105              | 0.270                |
| City 2 | 38                      | 1406            | 0.341 <sup>a</sup>  | 0.237                |
| City 3 | 44                      | 1892            | 0.389 <sup>b</sup>  | 0.296                |
| City 4 | 31                      | 930             | 0.238               | 0.243                |

<sup>a</sup> significant at 0.05;

<sup>b</sup> significant at 0.01.

ships are present in the data. As noted above, strong relations are those relations with values of 3 or 4 on the 0 to 4 scale. If systematic errors are present there should be significant positive correlations for cities within each network, indicating that organizations with larger numbers of strong relations will also have larger numbers of confirmed relations. Table 5 presents the findings. The correlation patterns are similar across cities and networks. Most are positive as expected, but there are only two significant correlations, both in the information network. City 1 presents an anomaly, where a small negative correlation appears in the information network. Although these results are not definitive, they offer some support for the hypothesis that the content of the relation influences the responses of the representatives. There is systematic error here, but these errors appear to be rather inconsequential when compared with those reported in cognitive network studies.

An additional question is of interest when assessing systematic errors in IO networks: Does the confirmation rate vary at different levels of relational strength? For each respondent a proportion can be computed expressing the number of times for a given response level the relationship is confirmed.<sup>12</sup> This is accomplished by first building a confirmation matrix based on the 0,1 adjacency matrix, and its transpose, as described above where a 1 in the confirmation matrix represents a confirmed relationship (*i.e.*  $i, j = j, i = 1$ ). The 0,1 confirmation matrix is then compared with the original matrix containing the 0 to 4 response values. Four variables are created for each

<sup>12</sup> We would like to thank an anonymous reviewer for suggesting this method.

respondent representing the total number of confirmed responses within each response category.<sup>13</sup> Additionally, for each respondent, a count was created for the number of times a particular response level was assigned to the relationship. Using these two totals *for each response level* the proportion of the responses that were confirmed relative to the number of reported relationships can be computed. By combining the confirmation variables for each city and dividing by the respective count totals, a proportion of confirmed relationships within each response level is provided for each network. These data are reported in Table 6.

Table 6 provides the number of times each response level was assigned by a respondent to describe a relationship with another organization, the number of times a relationship was confirmed at each response level, and the percentage of the relationships confirmed at each level. Results for both networks are reported by combining all city totals.<sup>14</sup> For both networks, the most often used response categories are on the low end of the scale, particularly the responses of 1. The lowest confirmation percentages consistently appear in the 1 response category. It is of interest to investigate the 1 responses in terms of how the alters are referring to these relationships. If most alter responses are 3, or 4 on the high end of the scale, (i.e. then these errors may signal total inability of respondents to accurately recall weak relationships). Of the 1 responses that were not confirmed in the coordination network, 64% of the alter organizations responded with a zero, whereas only 17% of the alter organizations responded with either a 3 or a 4 (e.g. on the high end of the scale). Relatively speaking then, the preponderance of errors does not reflect a great discrepancy in the perceived intensity of the relationship.

The most significant finding, however, is that the stronger the relationship the more likely it is to be confirmed. Overall for both

<sup>13</sup> The zero response category is not included.

<sup>14</sup> The authors have the results for each city individually which were consistent with the overall results in all response categories. For example, the ranges for the percent of confirmed relationships for the 1 to 4 response categories in the information network are: 47–52, 59–80, 75–87, 54–88, respectively. And in the coordination network the ranges are: 41–59, 50–73, 83–91, 76–88, respectively. The one major anomaly is the 54% of confirmed relationships in City 1 for response category 4 in the information network. If this figure is left out, the total percent confirmed for this response level would be 84%. We are at a loss in explaining why this figure is so low. As an aside the percent of no relationships (i.e. 0) that were confirmed are 73–77 and 61–72 for the information and coordination networks respectively.



Table 6  
The number and proportion of confirmed dyadic relationships within each response category for two networks and four cities

| Response level | Information network      |                                   |                                       | Coordination network     |                                   |                                       | Overall                  |                                   |                                       |
|----------------|--------------------------|-----------------------------------|---------------------------------------|--------------------------|-----------------------------------|---------------------------------------|--------------------------|-----------------------------------|---------------------------------------|
|                | Number of times assigned | Number of confirmed relationships | Percentage of confirmed relationships | Number of times assigned | Number of confirmed relationships | Percentage of confirmed relationships | Number of times assigned | Number of confirmed relationships | Percentage of confirmed relationships |
| 1              | 1054                     | 511                               | 48                                    | 1095                     | 564                               | 52                                    | 2149                     | 1075                              | 50                                    |
| 2              | 658                      | 431                               | 66                                    | 791                      | 507                               | 64                                    | 1449                     | 938                               | 65                                    |
| 3              | 314                      | 249                               | 79                                    | 437                      | 362                               | 83                                    | 751                      | 611                               | 81                                    |
| 4              | 260                      | 201                               | 77                                    | 302                      | 245                               | 81                                    | 562                      | 446                               | 79                                    |

networks approximately eight out ten relationships that were viewed as strong (i.e. 3 or 4) by the ego organization was confirmed by the alter, but only one-half of the very weak (i.e. 1) relationships were confirmed. While the percentages increase significantly between the response levels of 1–2, and 2–3, there is a slight decrease between the levels of 3 and 4. Given the large increases on the lower end of the scale we suspect that these represent statistically significant differences between the weak and strong relations. By using a difference of proportions test, we can verify the assumption. Testing the difference between the 2 response category and the 4 response category can establish whether the confirmation proportions in the weak relations categories (i.e. 1 and 2) are different than the strong relationship categories (i.e. 3 and 4). If this test is significant, then based on the relative sizes of the other proportions we can conclude that the confirmation percentages in response Categories 1 and 2 are significantly lower than in Categories 3 and 4 since the category percentages in response Categories 1 and 3 represent even larger differences than that between 2 and 4. A very significant Z score of 19.39 was obtained between Categories 2 and 4 providing evidence that the confirmation percentages are significantly lower in the weak response categories.<sup>15</sup>

These results are similar to other cognitive network data which reveals systematic errors resulting from the inclusion of relationships that are intensely connected (Bernard *et al.* 1980, 1982; Killworth and Bernard 1979; Freeman and Romney 1987; Hammer 1985).

## 6. Discussion

Two major concerns motivated this study. One is that the quality of network data gathered from organizational representatives is unknown. Because past studies have shown that people are unable to recall their contacts accurately, some researchers have suspected that the subjective nature of these data could create data quality problems. When using organizational representatives as respondents the recall problem is compounded because they are not only responding based on their own knowledge of their organization's relationships, but also

<sup>15</sup> The calculations are based on the proportions and number of responses in Table 6, Overall columns.

rely on their ability to accurately recall them. A second concern is that past studies raised the suspicion that more intense relationships produce more accurate responses, and as a result introduce systematic errors in network data. Both of these issues have a particular salience for IO studies since network methods provide ways for gathering the types of data of concern to IO researchers. But because IO networks are usually large and 'bounded' to include both strongly and weakly connected organizations, these issues should be explored by IO researchers wanting to use a network methodology.

We first assessed the problem of data quality by calculating the percentage of confirmed relationships given by organizational representatives and by correlating their mutual responses in two networks assumed to have a high degree of symmetry: information and coordination. The findings suggest a relatively high degree of data quality with approximately seven out of ten relations being mutually cited. One reason given for such high degrees of confirmed relationships is that persons are more likely to provide reliable responses when asked about routine relationships. Further, the quality of the data given by organizational representatives is probably proportional to how carefully they are chosen and how well 'bounded' the system of study is specified.

Next we assessed the possibility of systematic error by correlating, for each respondent, the proportion of their relationships that are reported as being intense with the proportion of their links that were confirmed. In addition confirmation proportions in each response category were created according to the number of times a relation was reported and confirmed. If systematic errors do exist, a pattern of large proportions would be apparent in the higher response categories. The findings suggest that in these data systematic errors are present as other studies have found, but there does not tend to be as many as in interpersonal cognitive networks.

Although these findings are viewed as preliminary, they are promising. Using organizational representatives to provide network data on IO relationships seems to be a reasonable choice given the strengths of the method. The a priori bounding of systems which include by necessity relationships known to be varied in their intensity may also be an acceptable method given these findings. If the extent of error reported in these data is thought to be excessive, one can increase data quality by interviewing several respondents from the same orga-

nization, assuming that multiple respondents provide better estimates of the true relationship. This, however, will increase the cost of data collection. In order to know whether the benefits of multiple respondents or other alternative strategies for increasing data quality outweigh the costs, more comprehensive studies of these issues have to be completed.

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